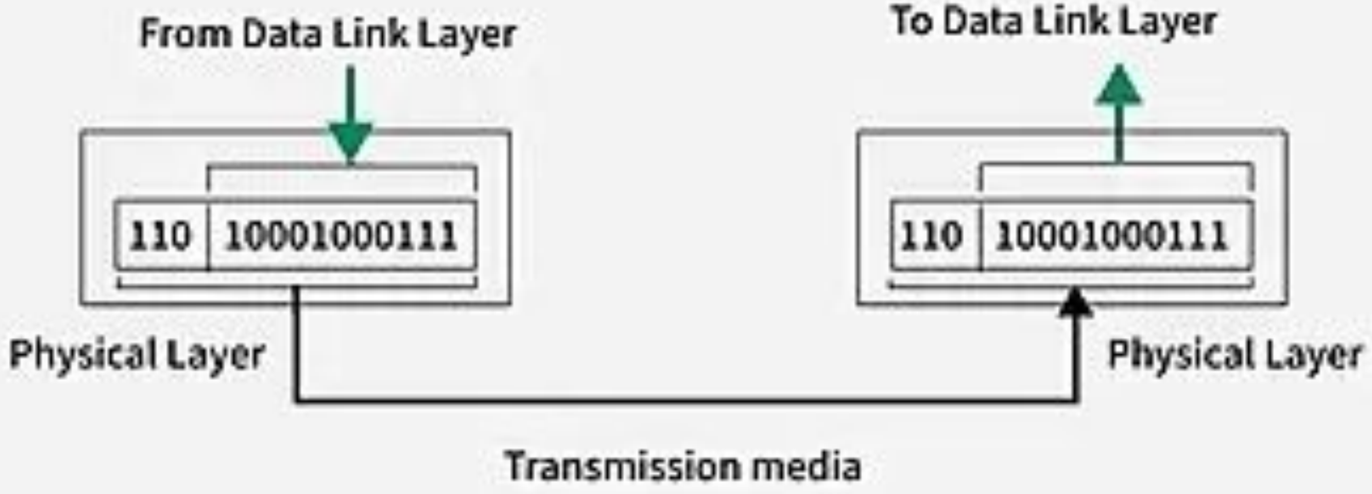
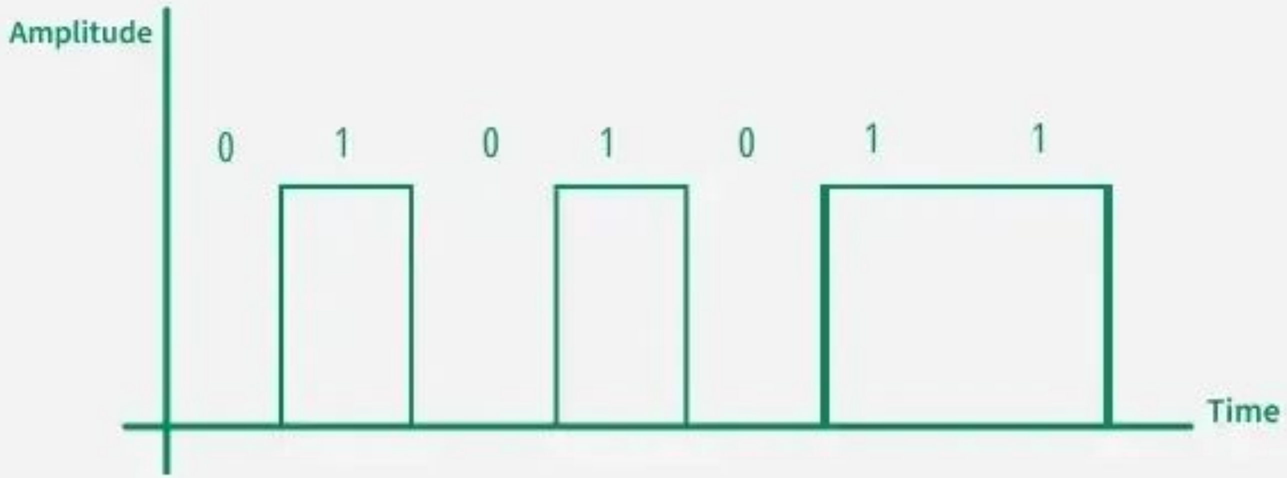
PHYSICAL LAYER

1. Bit-by-Bit Transmission



2. Encoding and Decoding

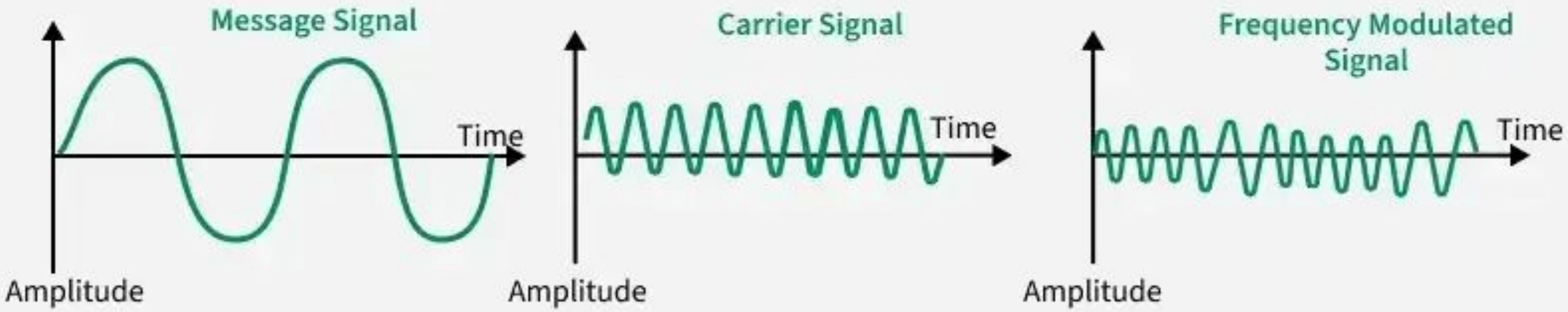


3. Signal Transmission

It is responsible for converting data into signals (analog or digital) for transmission and decoding these signals at the receiver's end.

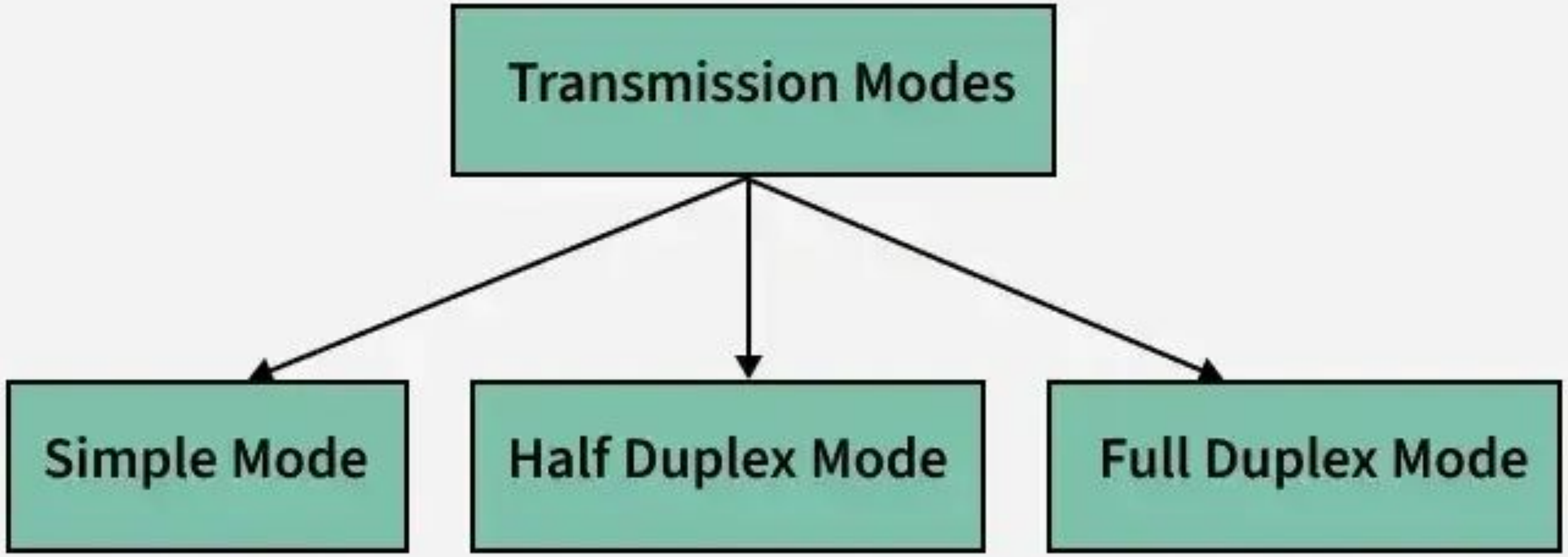
4. Modulation and Demodulation

Modulation is the process of modifying a carrier signal's properties (amplitude, frequency, or phase) for transmission over a communication medium.



5. Transmission Modes

Physical Layer determines the direction of data flow.

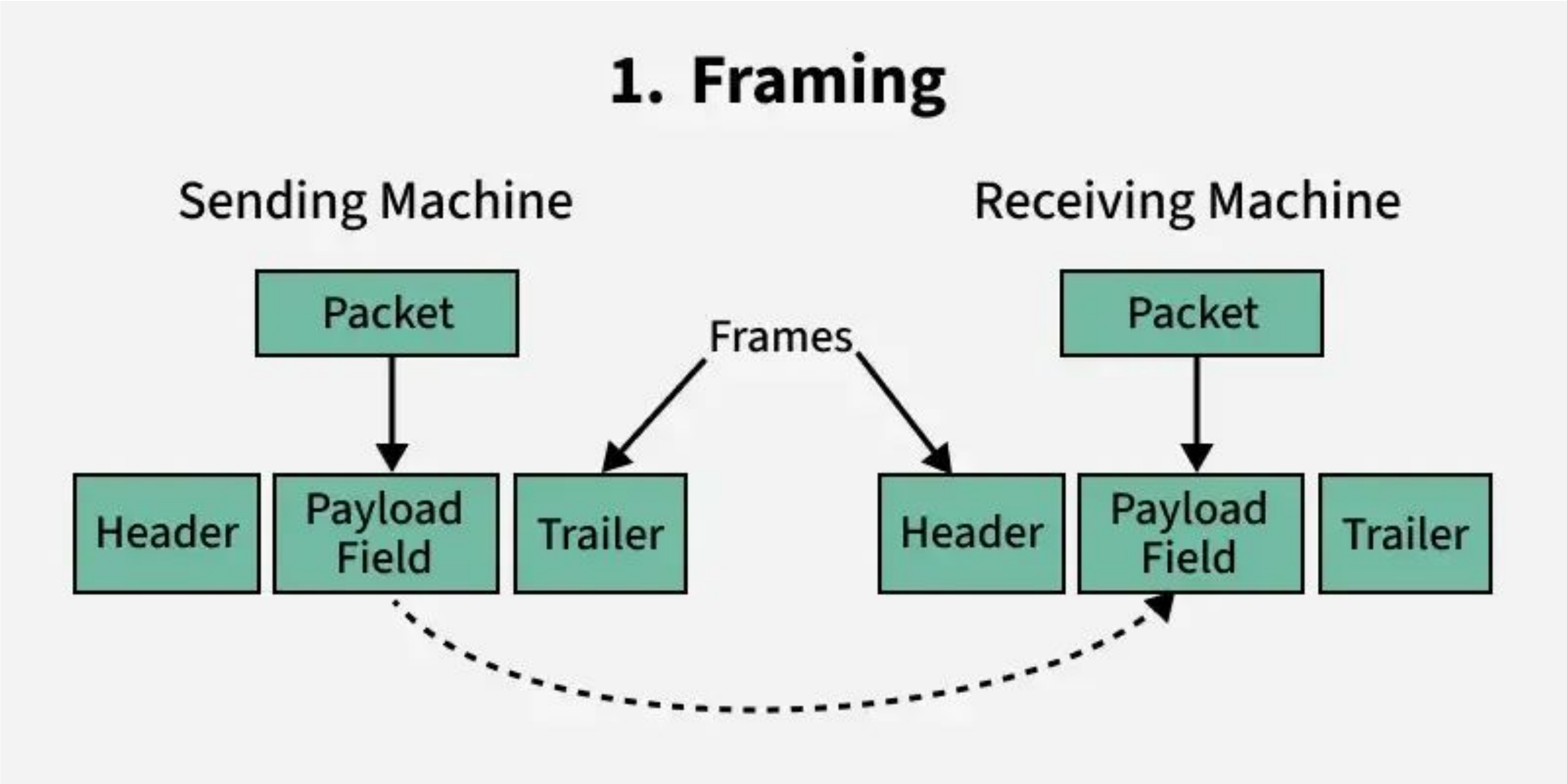


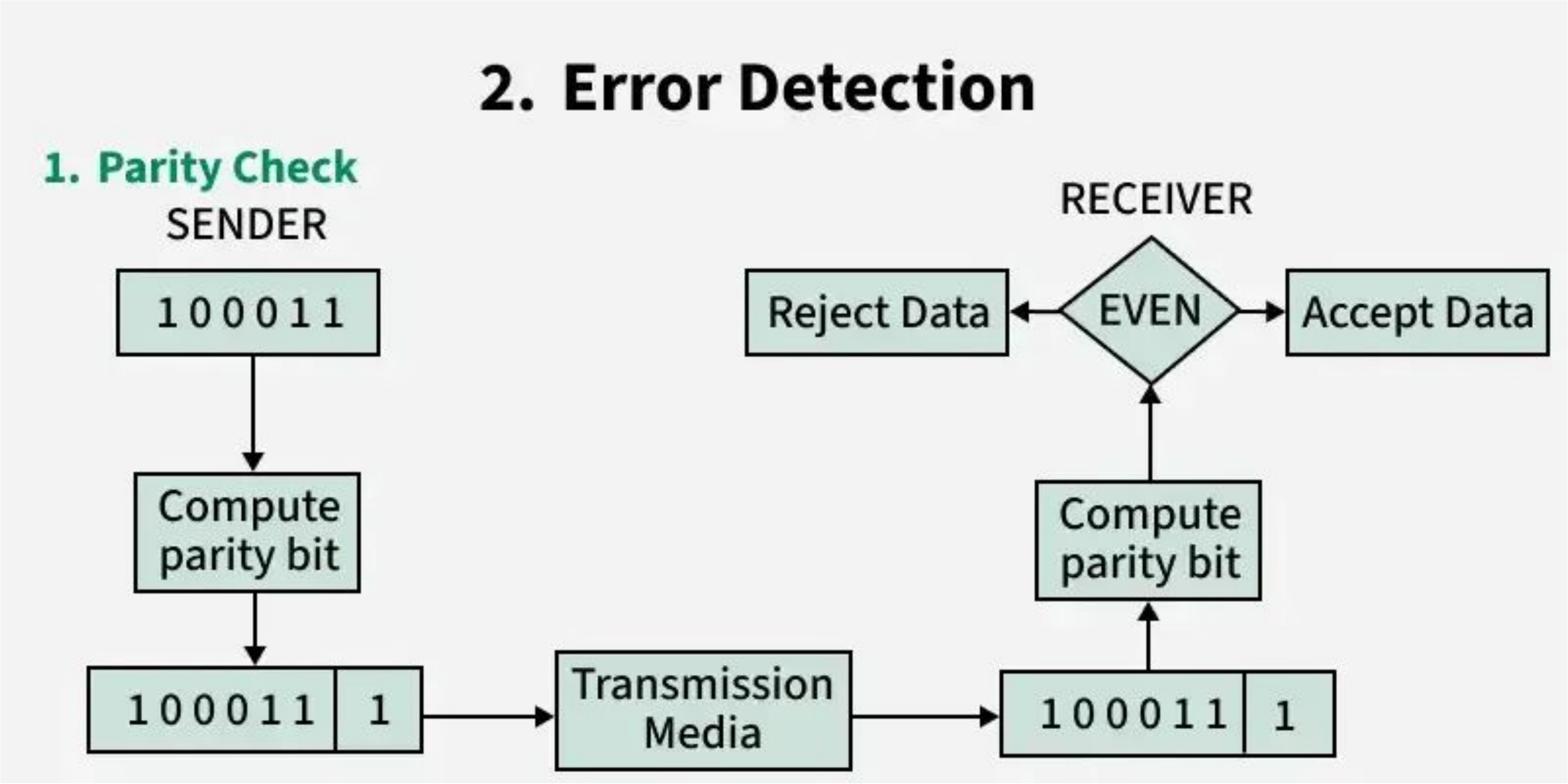
6. Data Rate Control

Physical layer controls the speed of data transmission so that sender and receiver use the same data rate.

- Prevents buffer overflow
- Reduces data loss
- Keeps transmission smooth

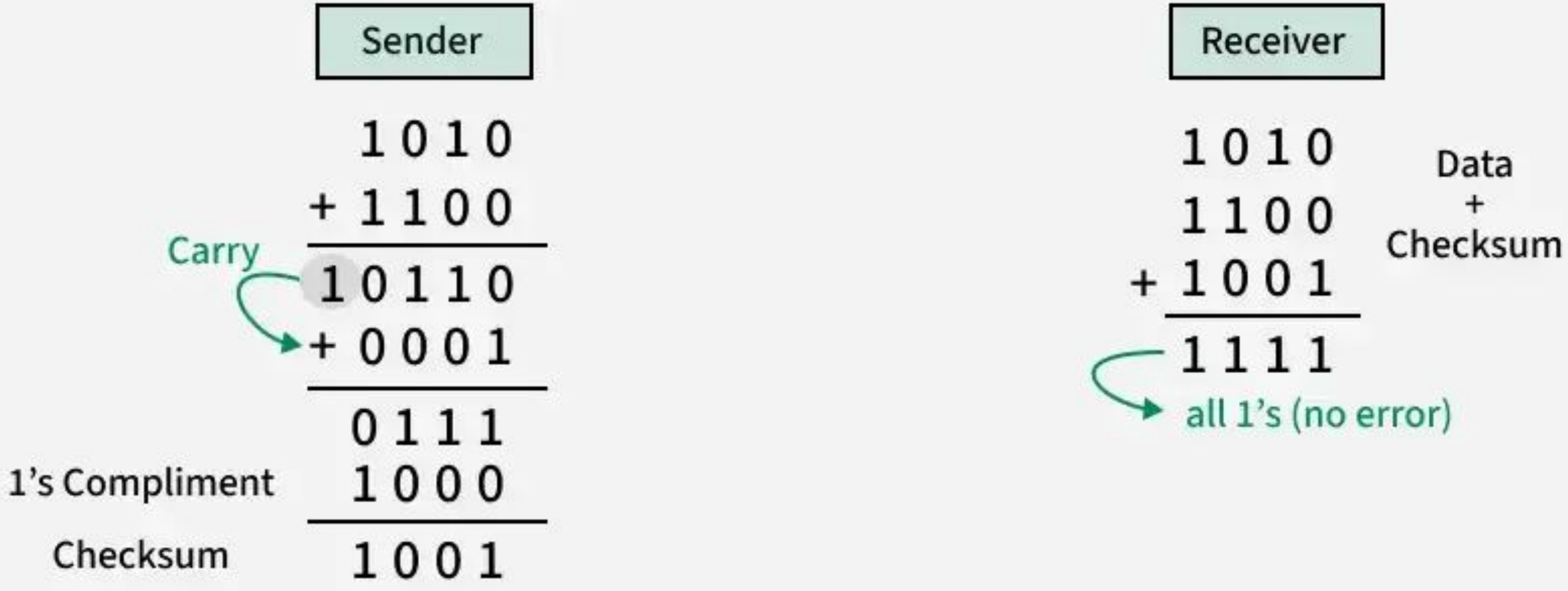
DATA LINK


1. Framing
Sending Machine
Receiving Machine
Packet
Frames
Packet
Header
Payload Field
Trailer
Header
Payload Field
Trailer


2. Error Detection
1. Parity Check
SENDER
1 0 0 0 1 1
Compute parity bit
1 0 0 0 1 1
1
Transmission Media
1 0 0 0 1 1
1
Compute parity bit
RECEIVER
EVEN
Reject Data
Accept Data

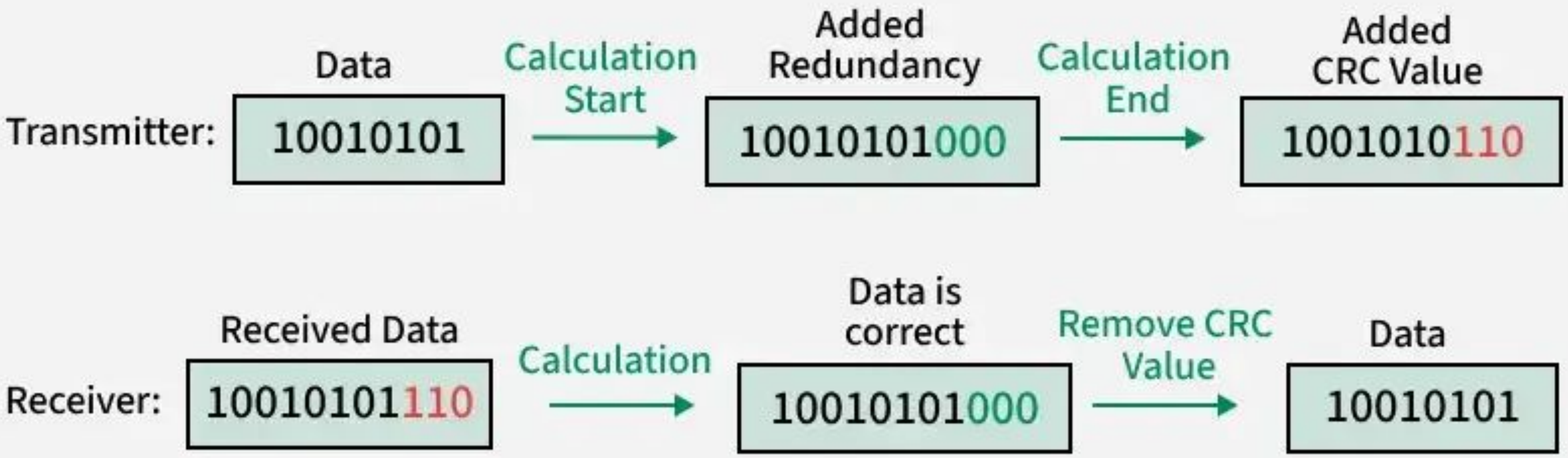
2. Error Detection

2. Checksum



2. Error Detection

3. Cyclic Redundancy Check



3. Error Correction

- **Automatic-repeat request (ARQ)**

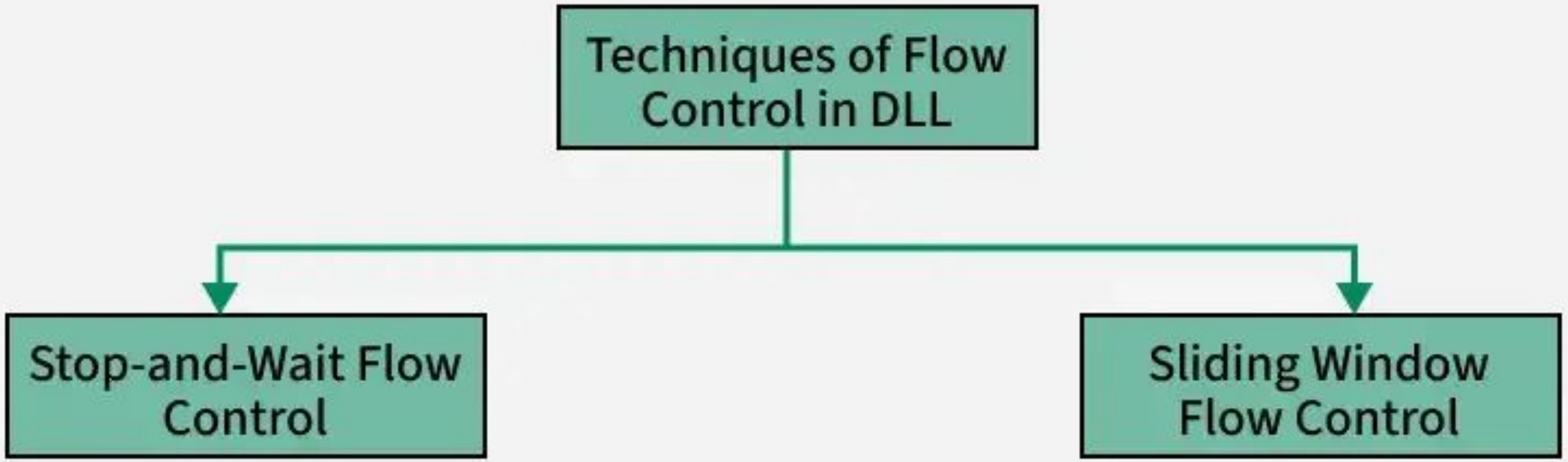
  ARQ is a feedback-based method where the receiver requests the sender to retransmit the data if errors are detected.

- **Forward error correction (FEC):**

  FEC is a method where the sender includes additional redundant data, allowing the receiver to detect and correct errors without retransmission.

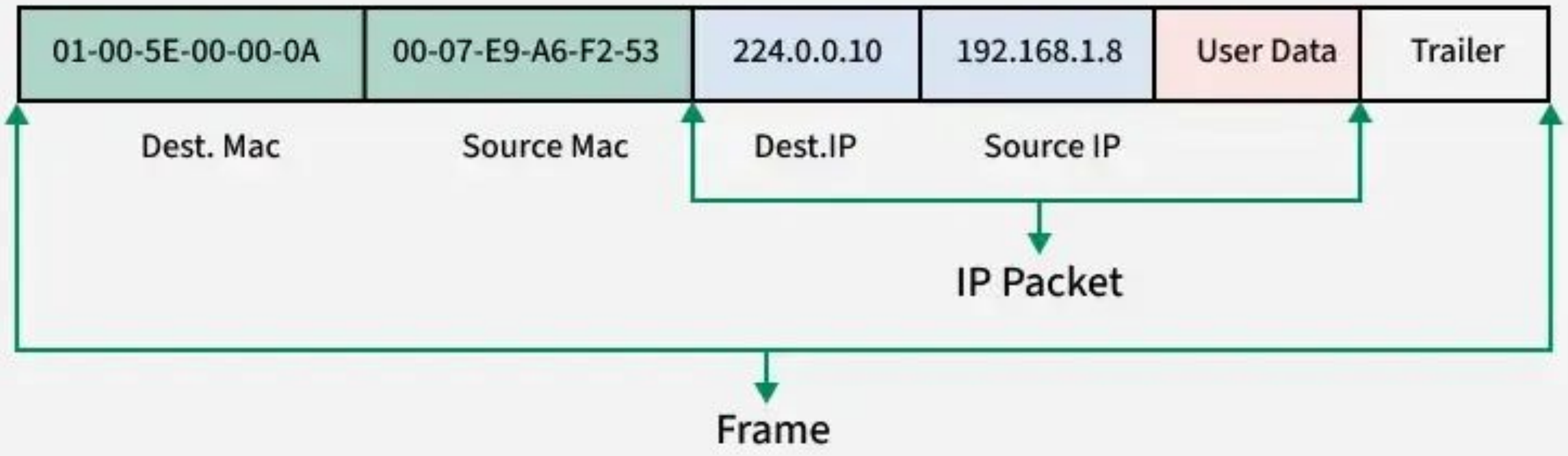
4. Flow Control

The Data Link Layer ensures flow control by synchronizing the sender's and receiver's speeds to prevent buffer overflow and frame loss.



5. Addressing

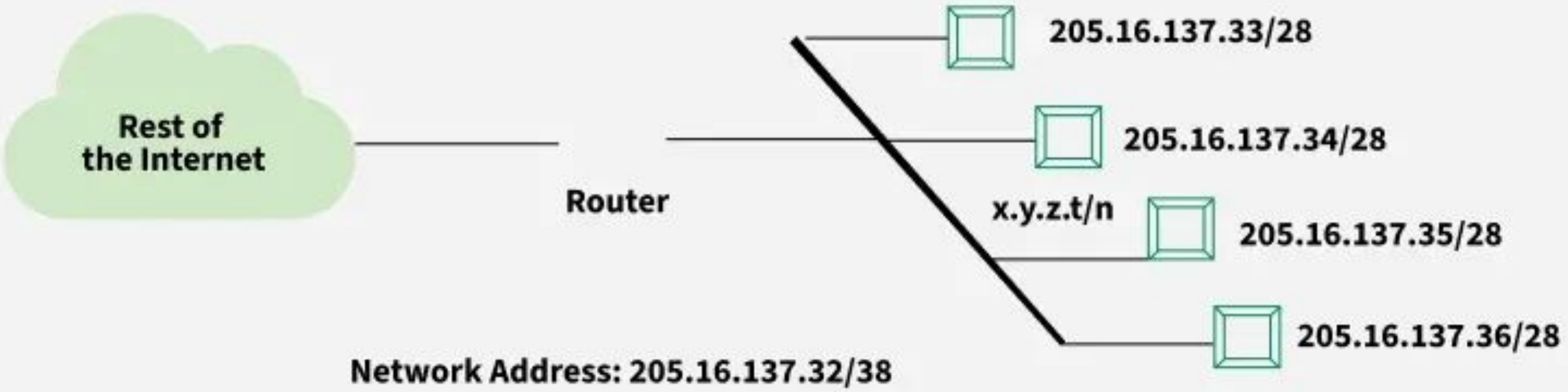
The Data Link Layer adds source and destination MAC addresses in the frame header for node-to-node delivery.

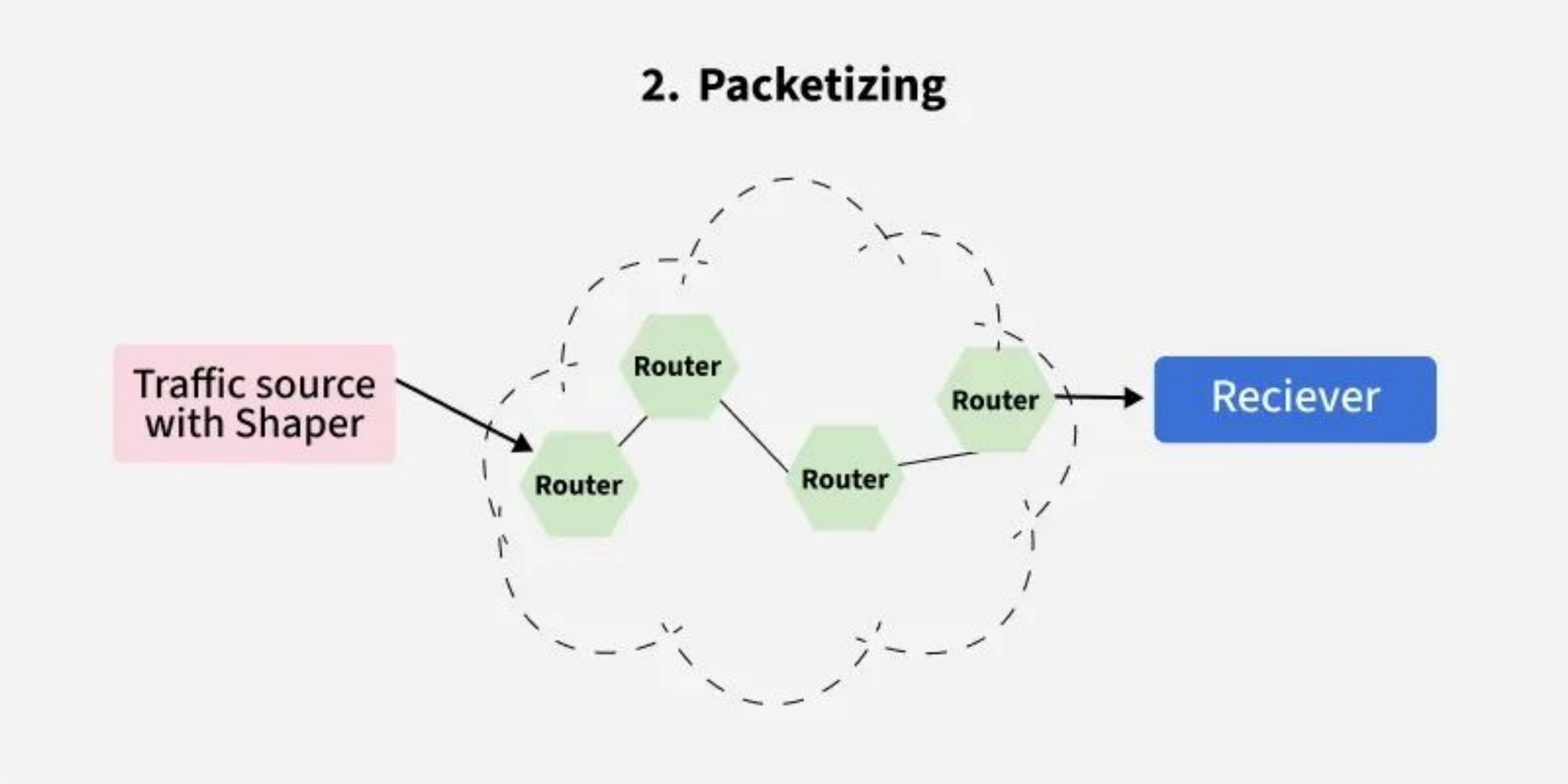


NETWORK

1. Assigning Logical Address

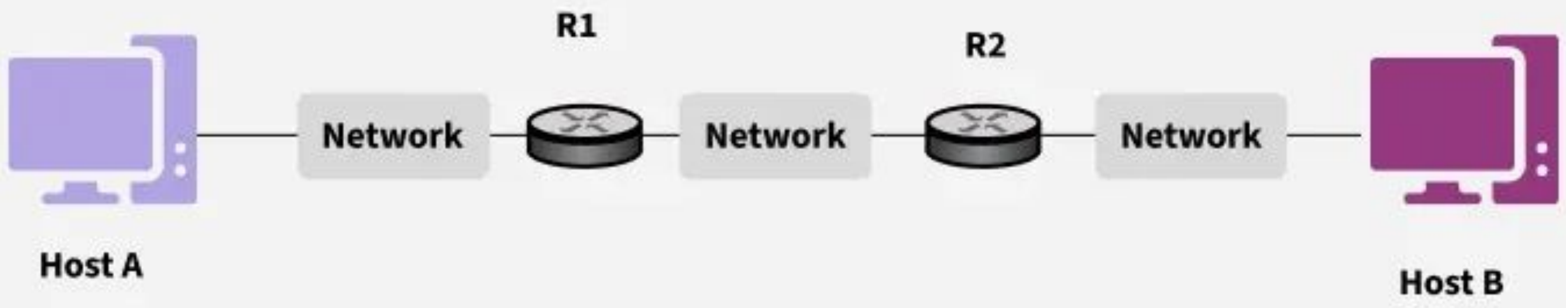
It assigns unique IP addresses to devices for identification across networks.




2. Packetizing
Traffic source with Shaper
Router
Router
Router
Router
Reciever

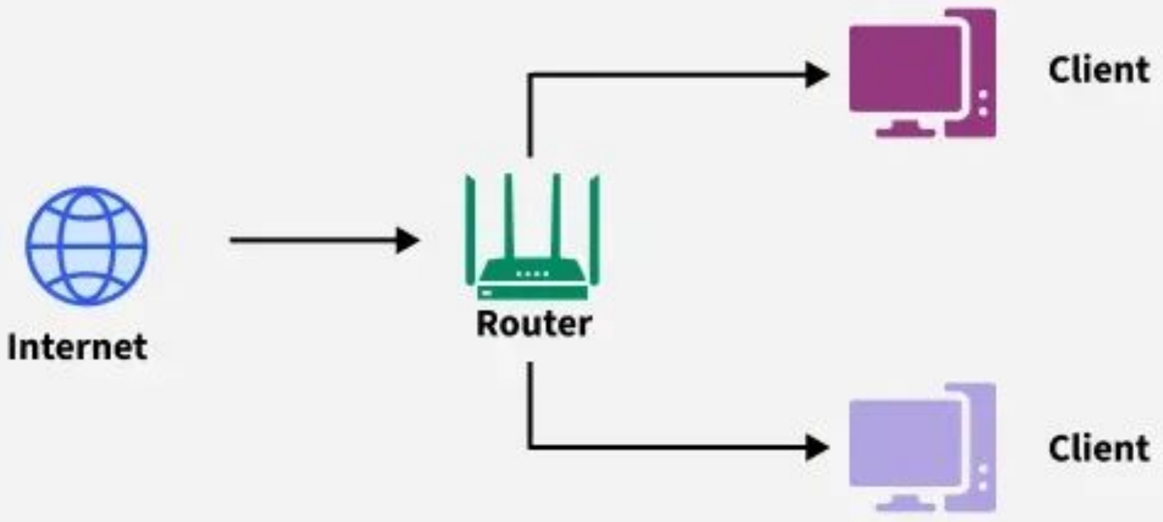
3. Host-to-host delivery

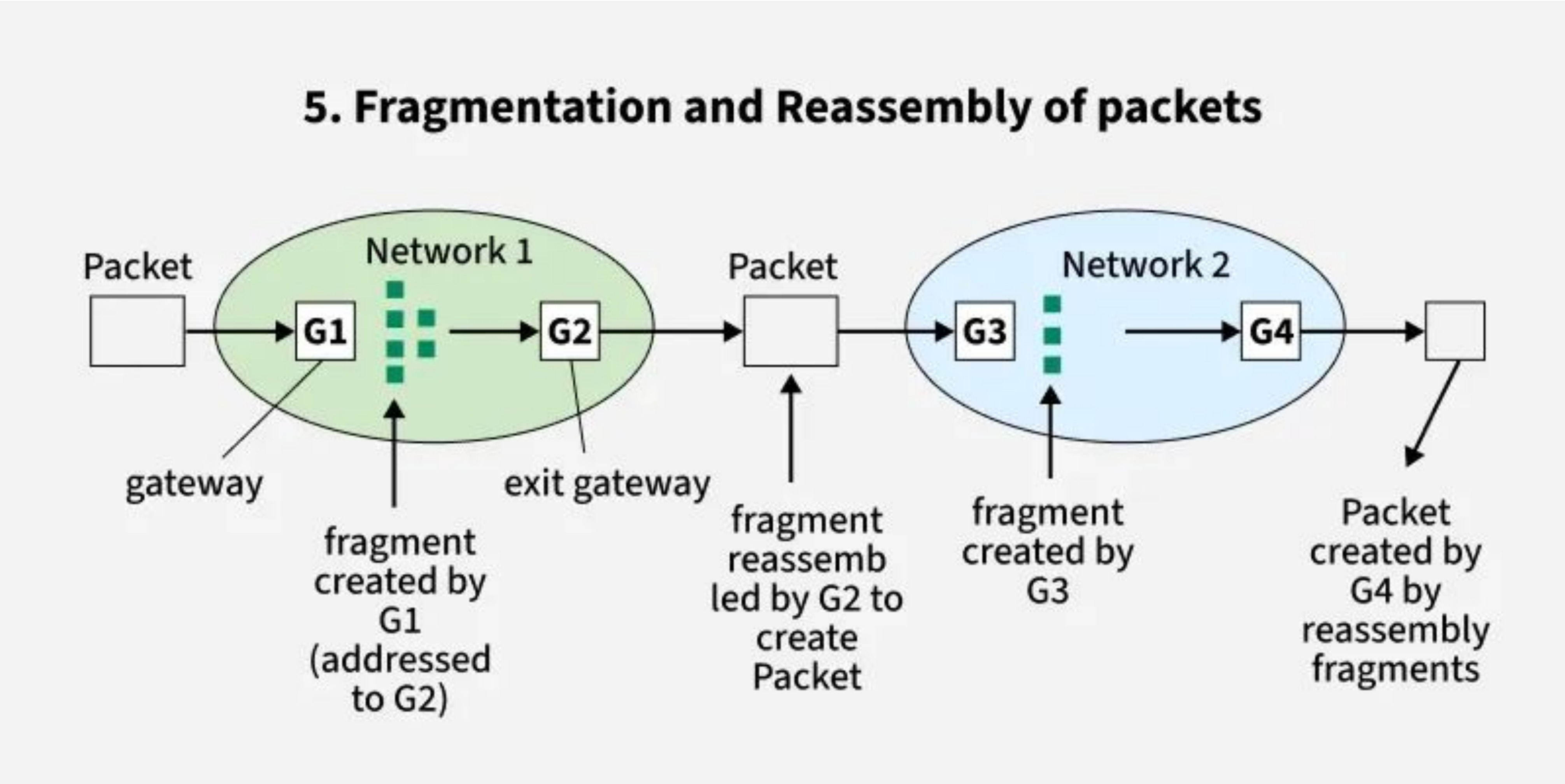
It ensures that data is reliably transmitted between two hosts across a network

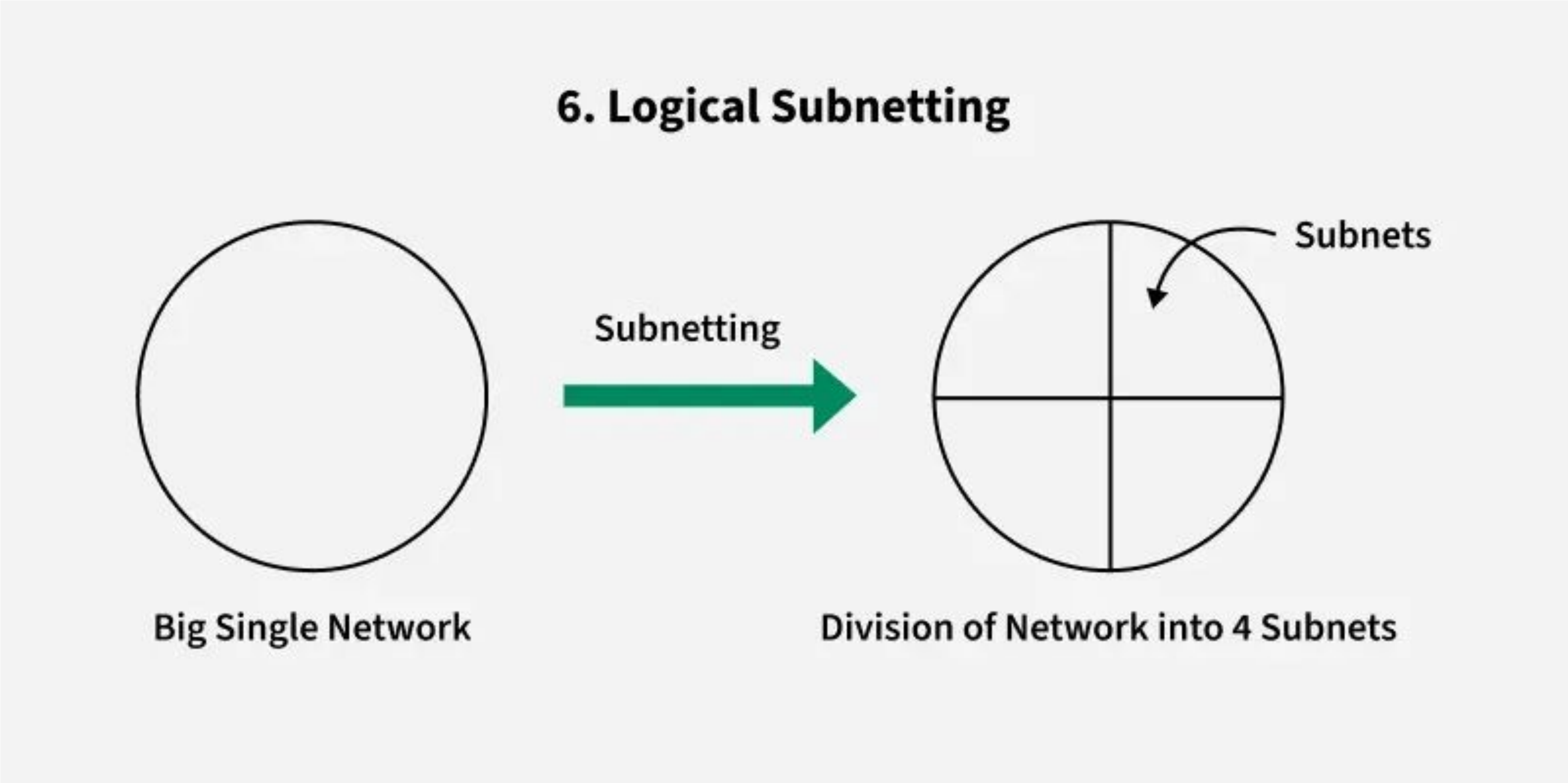


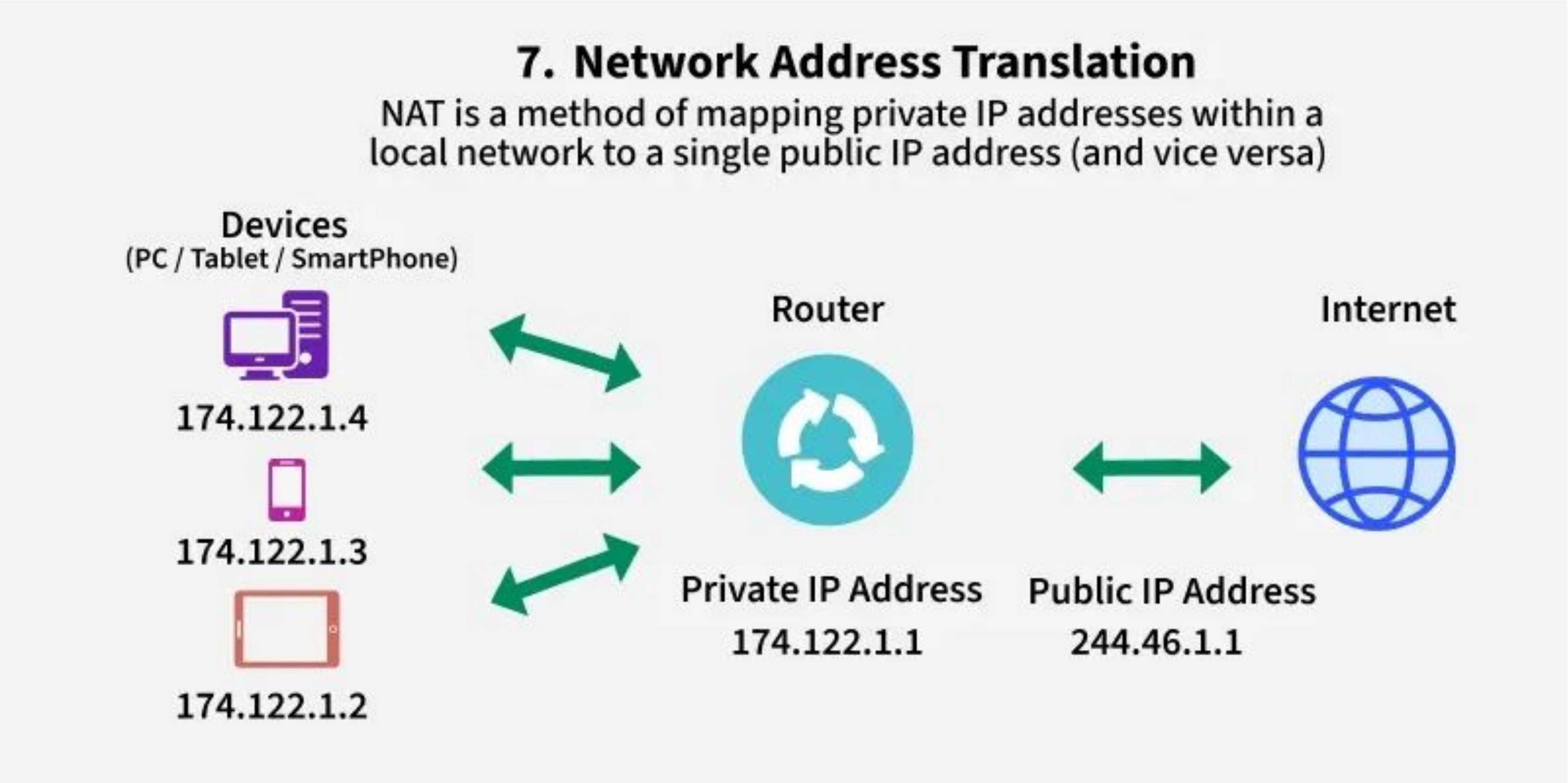
4.Forwarding

Forwarding moves packets through a router to the correct outgoing interface based on their destination.

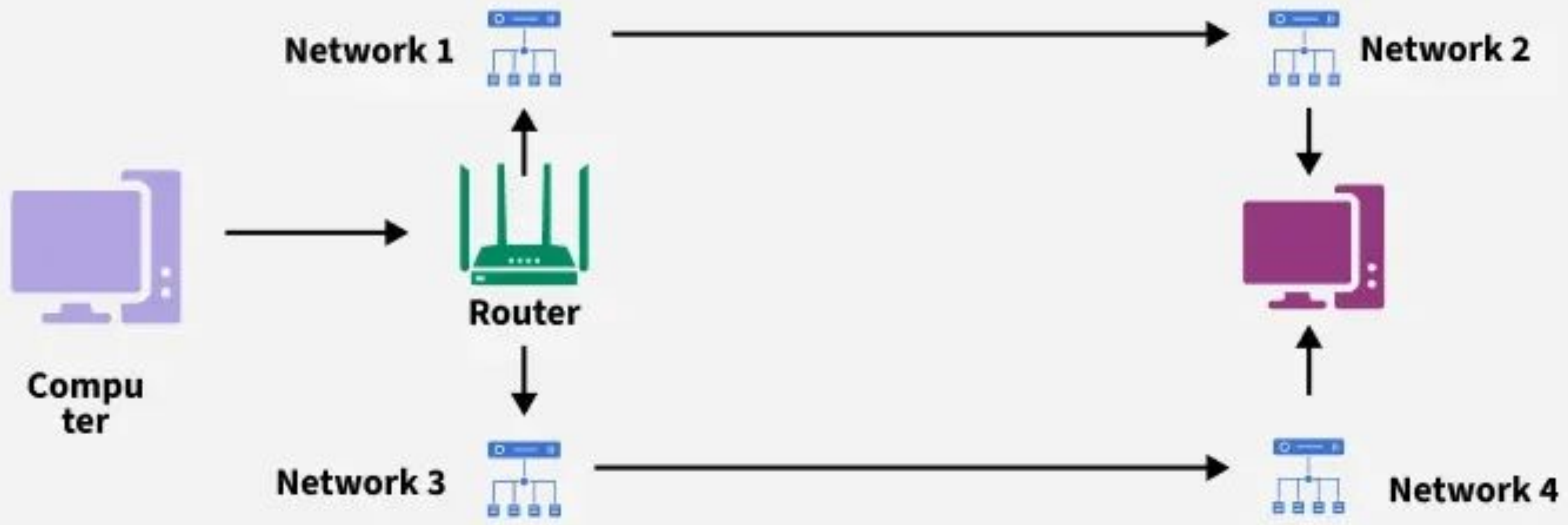



5. Fragmentation and Reassembly of packets
Packet
Network 1
G1
G2
Packet
Network 2
G3
G4
gateway
exit gateway
fragment created by G1 (addressed to G2)
fragment reassembled by G2 to create Packet
fragment created by G3
Packet created by G4 by reassembly fragments

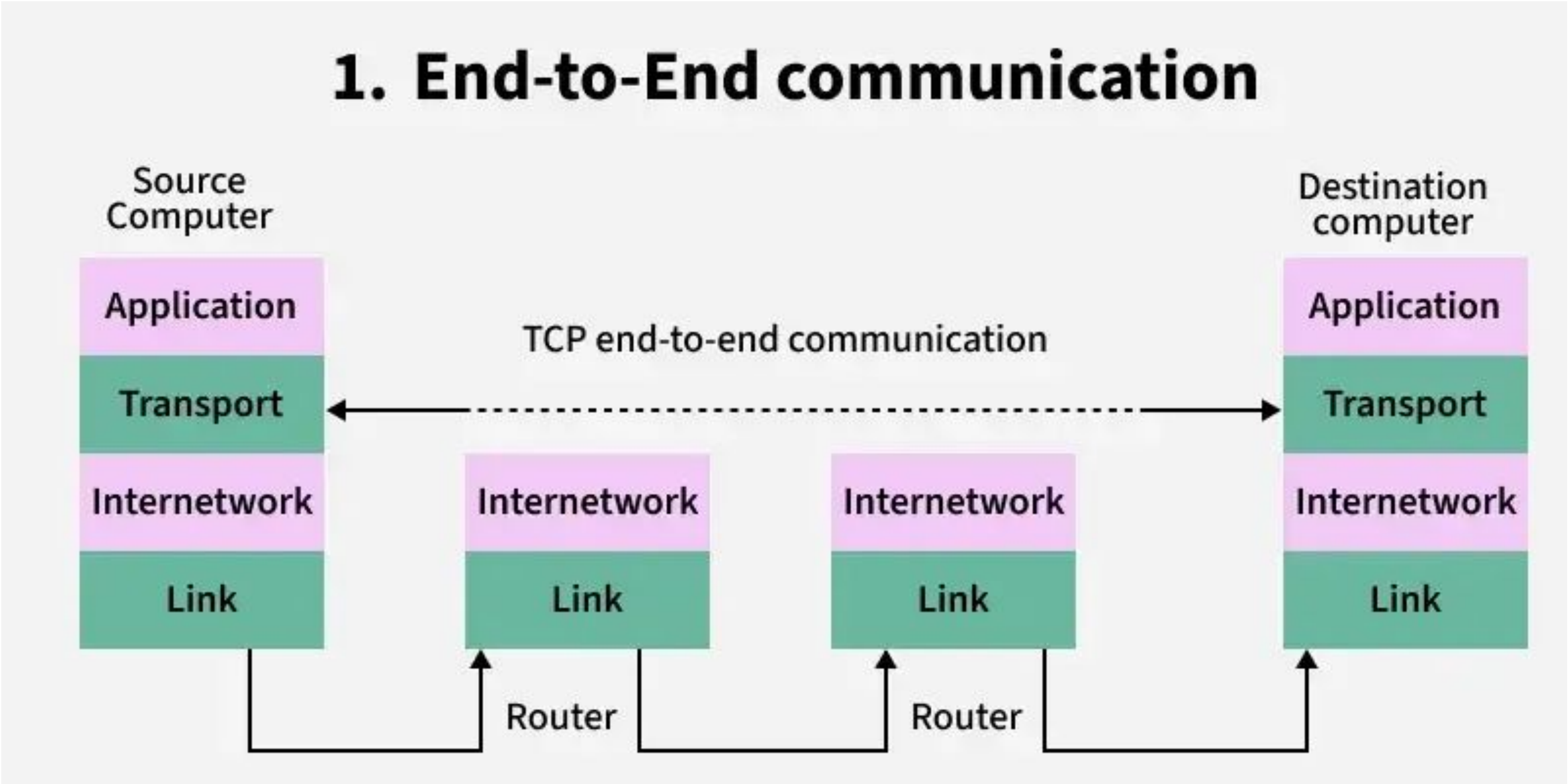

6. Logical Subnetting
Subnetting
Subnets
Big Single Network
Division of Network into 4 Subnets

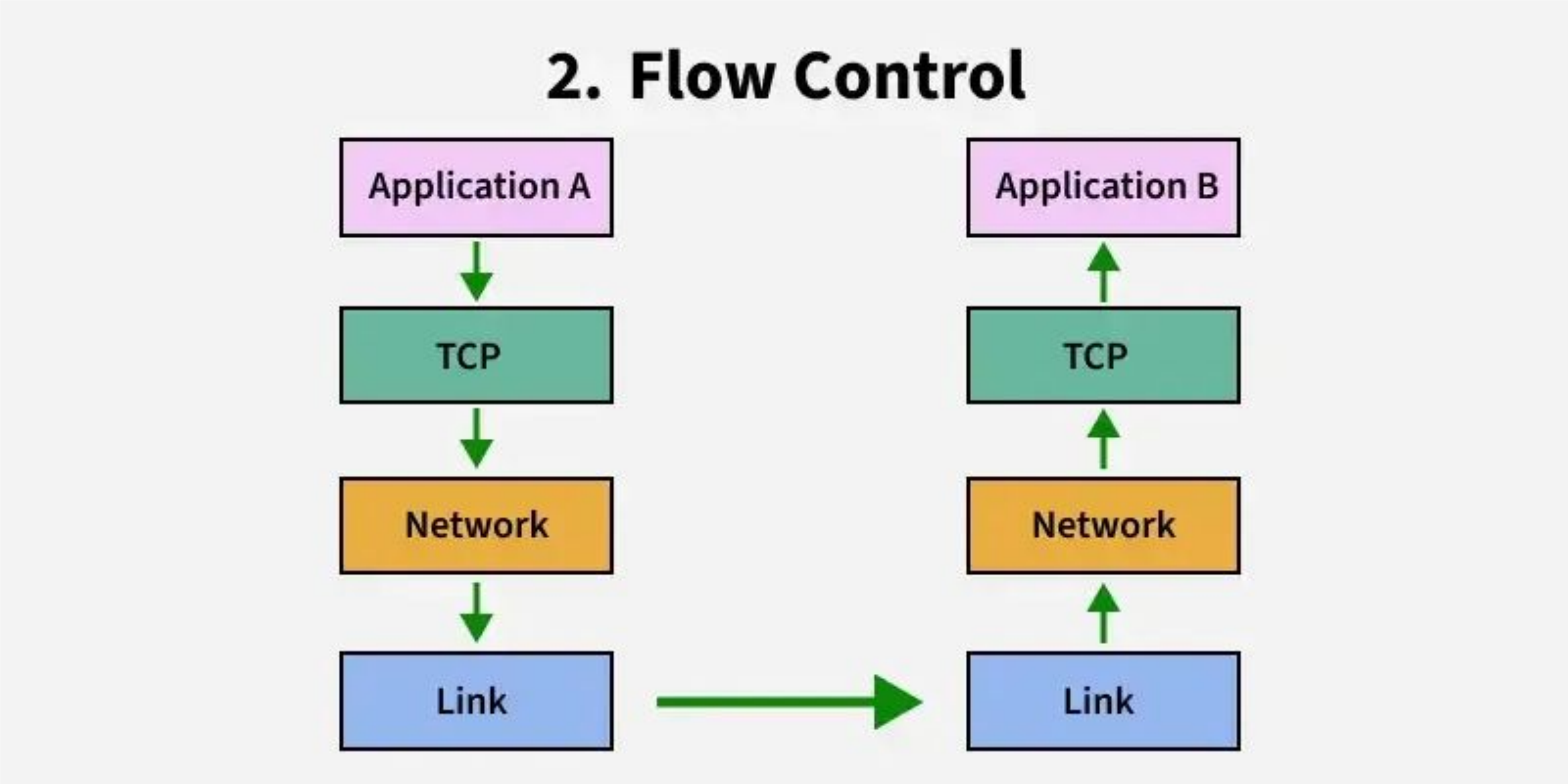

7. Network Address Translation
NAT is a method of mapping private IP addresses within a local network to a single public IP address (and vice versa)
Devices
(PC / Tablet / SmartPhone)
174.122.1.4
174.122.1.3
174.122.1.2
Router
Private IP Address
174.122.1.1
Public IP Address
244.46.1.1
Internet

8. Routing



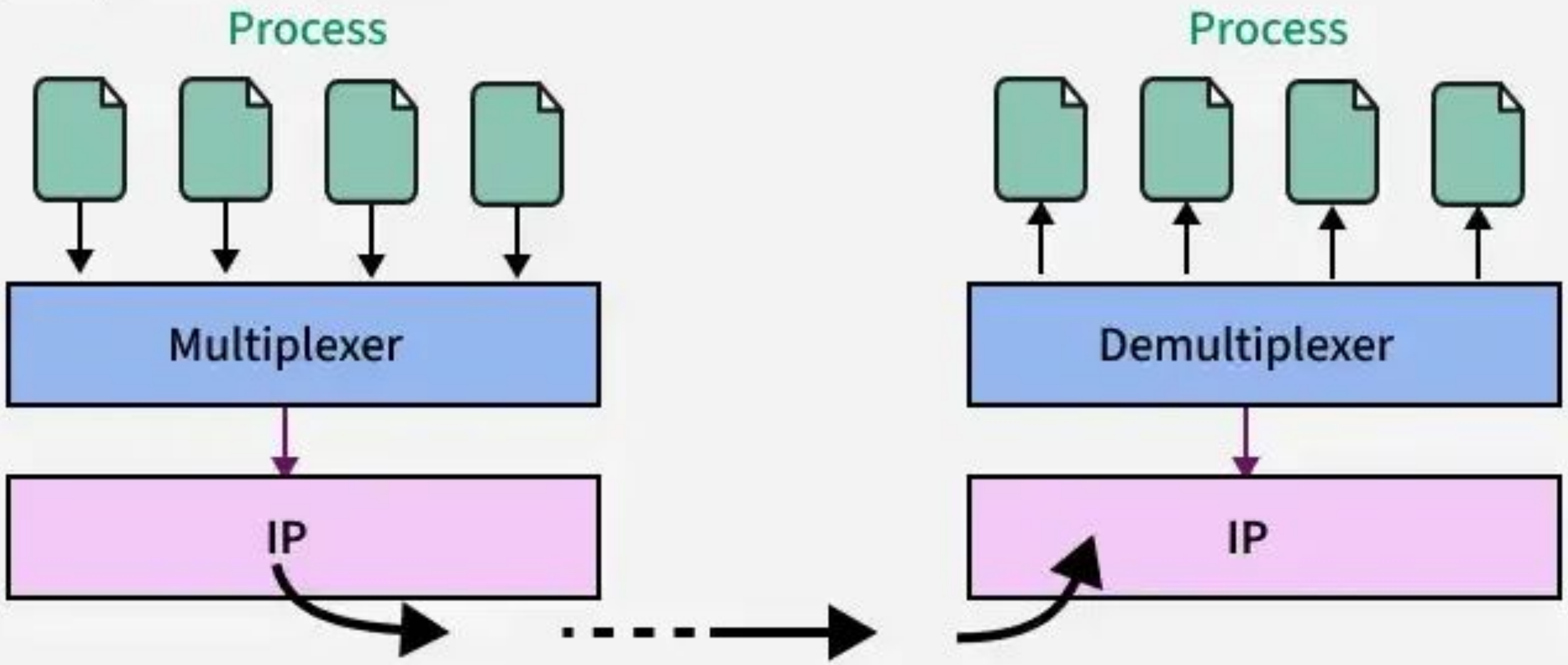
TRANSPORT


1. End-to-End communication
Source Computer
Destination computer
Application
Transport
Internetwork
Link
TCP end-to-end communication
Internetwork
Link
Internetwork
Link
Application
Transport
Internetwork
Link
Router
Router

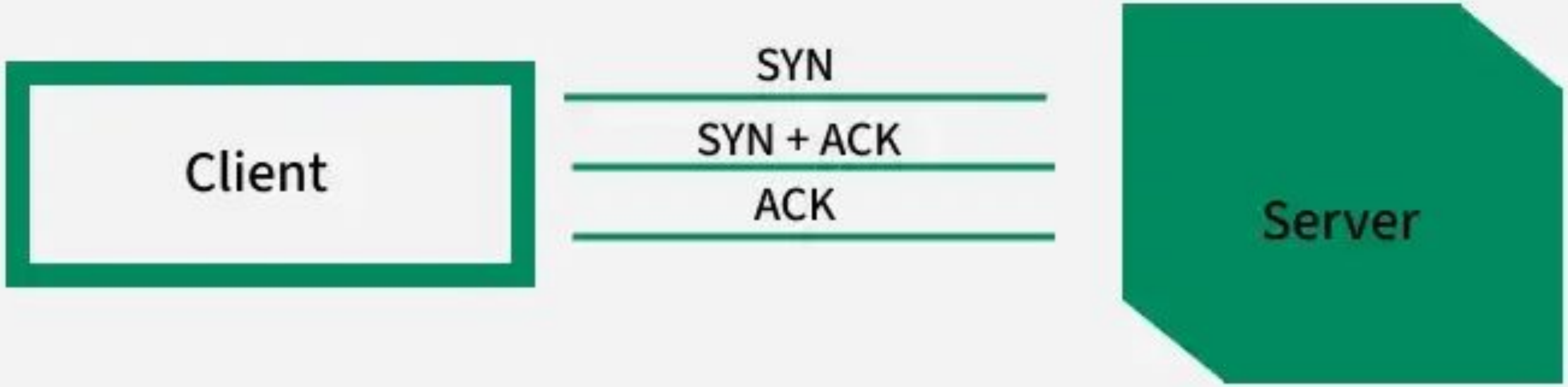

2. Flow Control
Application A
TCP
Network
Link
Application B
TCP
Network
Link

3. Multiplexing and Demultiplexing

It manages multiple data streams on the same network connection by assigning unique ports.

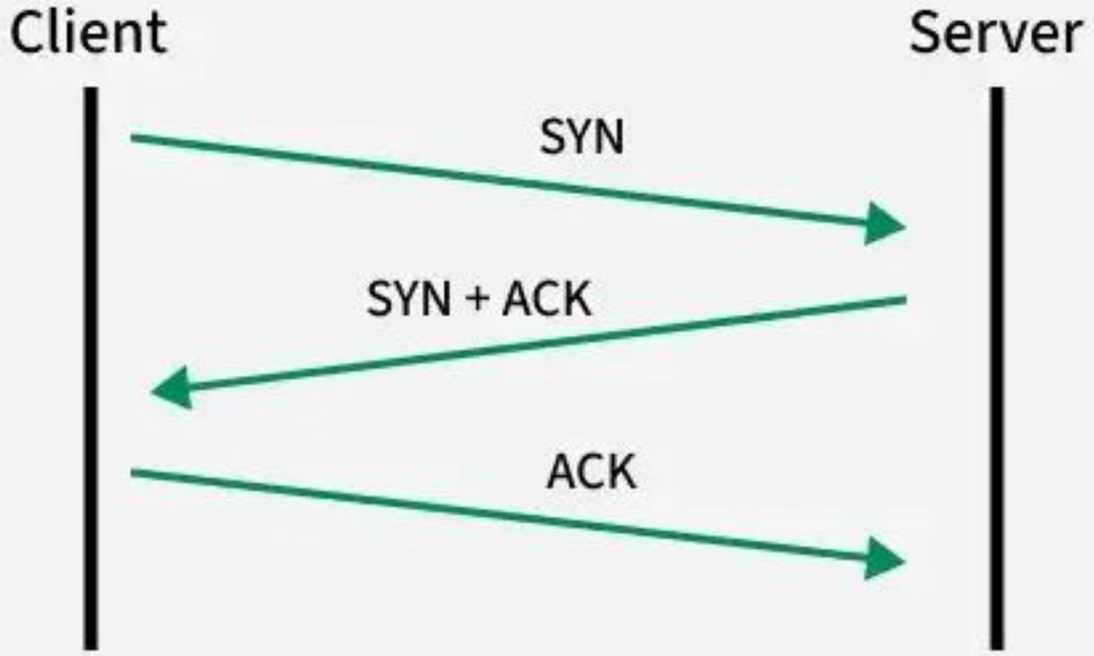


4. Connection Establishment



TCP 3 Way Handshake

5. Connection Termination



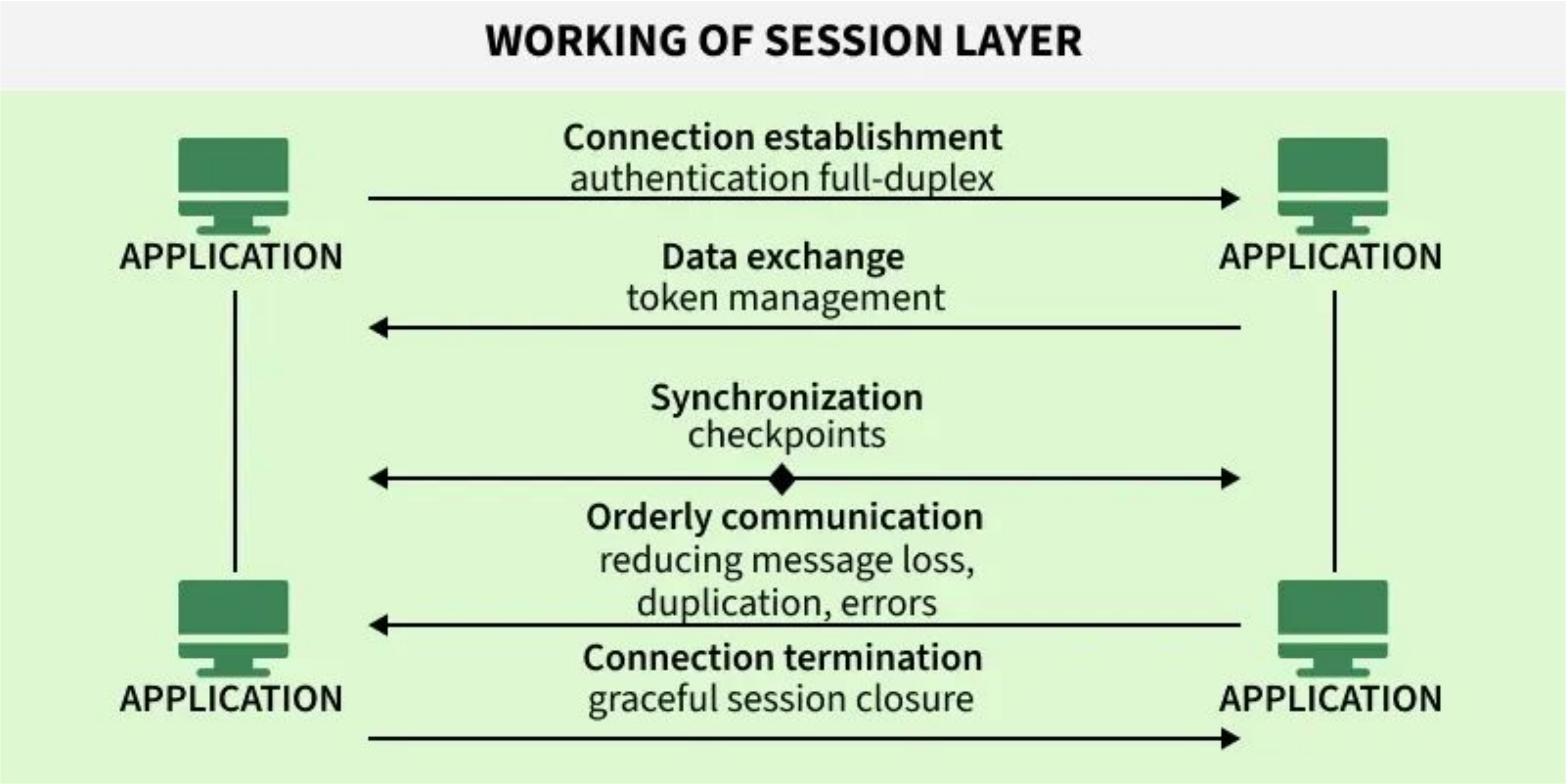
6. Reliable Data Delivery

It ensures all data is delivered correctly, in sequence, and without duplication (in protocols like TCP).

7. Quality of Service (QoS) (Optional)

It ensures priority and performance for specific types of data, like video calls or file transfers.

SESSION


WORKING OF SESSION LAYER
APPLICATION
Connection establishment
authentication full-duplex
APPLICATION
Data exchange
token management
Synchronization
checkpoints
Orderly communication
reducing message loss,
duplication, errors
APPLICATION
Connection termination
graceful session closure
APPLICATION

APPLICATION

1. Data Representation

It ensures that data is in a readable format for both the sender and receiver, including data translation, compression, and encryption.

2. Network Service Access

It provides applications with access to network services, enabling communication over a network.

3. Application Protocols

It supports protocols like HTTP, FTP, SMTP, and DNS that enable communication between applications.

4. Session Management

It manage establishment, maintenance, and termination of communication sessions between applications.